

ALEXIS JONO TAN

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EDUCATION

GEORGIA INSTITUTE OF TECHNOLOGY

B.S. in Computer Science, Concentration in Intelligence, Systems & Architecture

Atlanta, GA

Expected May 2028

- Relevant Courses: Data Structures & Algorithms, Object-Oriented Programming with Java, Computing with Python

GWINNETT SCHOOL OF MATHEMATICS, SCIENCE, AND TECHNOLOGY

GPA: 4.43 | PSAT: 1520/1520 | SAT: 1560/1600

Atlanta, GA

Aug 2022 - May 2026

- Activities: Math Team, Computer Science Club, Mu Alpha Theta, National Honors Society
- Awards: National Merit Finalist (2025), American Invitational Mathematics Exam (AIME) Qualifier (2025)

EXPERIENCE

GEORGIA TECH RESEARCH INSTITUTE

Research Intern, Information and Communications Lab

Atlanta, GA

Jun 2026 - July 2026

- Co-developed a Python/Flask full-stack web application within a 3-person Agile team, collaborating with a 10+ member hardware team to process and visualize live environmental sensor data from the Ocmulgee River
- Built a dynamic dashboard using HTML/CSS/JS and Chart.js to visualize raw river metrics, enabling early flood-warning capabilities, supporting both pre-disaster mitigation efforts and environmental health monitoring
- Engineered the backend data pipeline to efficiently route real-time telemetry CSV data to the frontend using REST APIs
- Partnered with the Ocmulgee National Park and Preserve Initiative through participatory design research, engineering data-accessibility tools to support cultural preservation and the establishment of Georgia's first national park

VALENT ASSET MANAGEMENT

Quantitative Analyst Intern, Metals

New York, NY

Jun 2025 - Apr 2026

- Engineered quantitative trading strategies for base, precious, and minor metals portfolios, leveraging time-series analysis and regression on Bloomberg price data to generate trading signals yielding an 80%+ success rate and 5-10% average PnL
- Built customized trading and backtesting programs using Python/Pandas to evaluate strategy performance, directly optimizing portfolio returns against benchmark metrics
- Synthesized 50+ pages of weekly commodities and macroeconomic research from major institutions (Goldman Sachs, Macquarie, BMO, Bloomberg) into concise briefs, streamlining the research pipeline for senior quantitative analysts

FREELANCE

Full-Stack Web Developer

Atlanta, GA

May 2025 - Aug 2025

- Architected and deployed custom web applications for two foodservice businesses, using Render and Git, independently designing Python/Flask backends and responsive HTML/CSS/JavaScript frontends to establish their online presence
- Drove digital customer engagement by optimizing site visibility and designed responsive UI/UX, successfully growing combined platform traffic from 0 to 200+ monthly active visitors
- Engineered secure B2B onboarding forms for credential intake and integrated a custom contact routing system, streamlining both customer catering inquiries and vendor onboarding

PROGRAMMING PROJECTS

EVALUATING CAROTID INTIMA-MEDIA THICKNESS USING MACHINE LEARNING

- One of top 50 research projects in Georgia chosen for presentation at Georgia Junior Science & Humanities Symposium
- Engineered machine learning and deep learning pipelines using TensorFlow to automatically measure carotid intima-media thickness from ultrasound scans, achieving 90%+ diagnostic accuracy
- Curated and open-sourced a comprehensive dataset of 2,000+ labeled medical ultrasound scans, enabling future research in computer vision and image segmentation tasks
- Authored a comprehensive 20+ page research paper detailing methodology and results, and delivered a technical presentation to a panel of academic researchers

TECHNICAL SKILLS

Languages: Python, Java, C++, HTML, CSS, JavaScript, TypeScript, SQL (PostgreSQL, MySQL)

Frameworks: Agile (SDLC), Unit Testing (JUnit), Git (GitHub, GitLab), Linux, Flask, React, Node.js, Express.js

AI & ML: PyTorch, TensorFlow, Scikit-learn, Keras, Pandas, Numpy, OpenCV, Feature Engineering, Model Evaluation

Other: Object-Oriented Programming (OOP), AI-Assisted Development, REST APIs, Jira, Time Series Analysis